



# Artificial Intelligence

## Tutorial : Setup a Python Environment for AI labworks

(or any other Python related JUNIA projects or courses)

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# Why Python ?

- Why is Python the best for Artificial Intelligence and Machine Learning?  
<https://www.tristatetechnology.com/blog/why-is-python-the-best-for-artificial-intelligence-and-machine-learning/>
- 8 Reasons Why Python is Good for AI and ML  
<https://djangostars.com/blog/why-python-is-good-for-artificial-intelligence-and-machine-learning/>
- **AI answer** Python is the best language for AI and ML due to its simplicity, readability, and vast ecosystem of libraries like TensorFlow, PyTorch, and Scikit-learn. Its easy syntax accelerates development and debugging, allowing researchers to focus on algorithms rather than code complexity. Python's active community, extensive documentation, and support for integration with other languages and tools also make it ideal for AI and ML tasks.

# Choose your IDE



**VSCode**

Local IDE

+ Github Copilot



**JetBrains PyCharm**

Local IDE



**Jupyter notebook**

Local Web



**Google Colab**

Cloud

# Python AI/ML/DL environment

Python packages

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                               |                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  python™                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Python                                                        | The interpreter (or the programming language)                                                                                                                                                                                                          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Conda                                                         | An environment manager (like <a href="#">virtualenv</a> , <a href="#">pyenv</a> , <a href="#">uv</a> , <a href="#">poetry</a> , ... ).<br>A command line tool.<br>Allows to install multiple versions of Python or different library sets and versions |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Anaconda<br>&<br>Miniconda                                    | A python distributions, Simplify package and virtual environment management<br><b>Anaconda = Python + Conda + A set of pre-installed useful data science libraries</b><br><b>Miniconda = Python + Conda</b>                                            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Jupyter                                                       | Web-based interactive development environment for code, data and notebooks.<br>Distributed as a python package (Depends on Python, but not on Anaconda).<br><b>Labworks, evaluations and exam will be done in Jupyter.</b>                             |
|  NumPy<br> learn<br> K +  TensorFlow<br> pandas<br> OpenCV<br>... | Numpy<br>Scikit-learn<br>Tensorflow<br>Keras<br>Pandas<br>... | Python packages for artificial intelligence, machine learning, deep learning, data science, scientific computing, visualization, machine vision, natural language processing, ...                                                                      |

| Feature           | Conda        | Venv                 | UV                      | Poetry               |
|-------------------|--------------|----------------------|-------------------------|----------------------|
| Environment Mgmt  | ✅            | ✅                    | ✅                       | ✅                    |
| Package Mgmt      | ✅<br>(Conda) | ❌ (uses<br>pip)      | ✅ (pip-<br>compatible)  | ✅ (Poetry)           |
| Non-Python Deps   | ✅            | ❌                    | ❌                       | ❌                    |
| Lockfiles         | ❌            | ❌                    | ✅ ( uv pip<br>compile ) | ✅<br>( poetry.lock ) |
| Speed             | 🕒 Slow       | ⚡ Fast<br>(envs)     | ⚡⚡⚡ Blazing<br>fast     | 🕒 Moderate           |
| CI/CD Optimized   | ❌            | ⚠️ (requires<br>pip) | ✅                       | ⚠️                   |
| Packaging/Publish | ❌            | ❌                    | ❌                       | ✅                    |
| Built-in Python   | No           | Yes (3.3+)           | No                      | No                   |

# IDE vs environment

- Don't confuse your python IDE with python environment
- You can configure multiple environments and install multiple IDE.

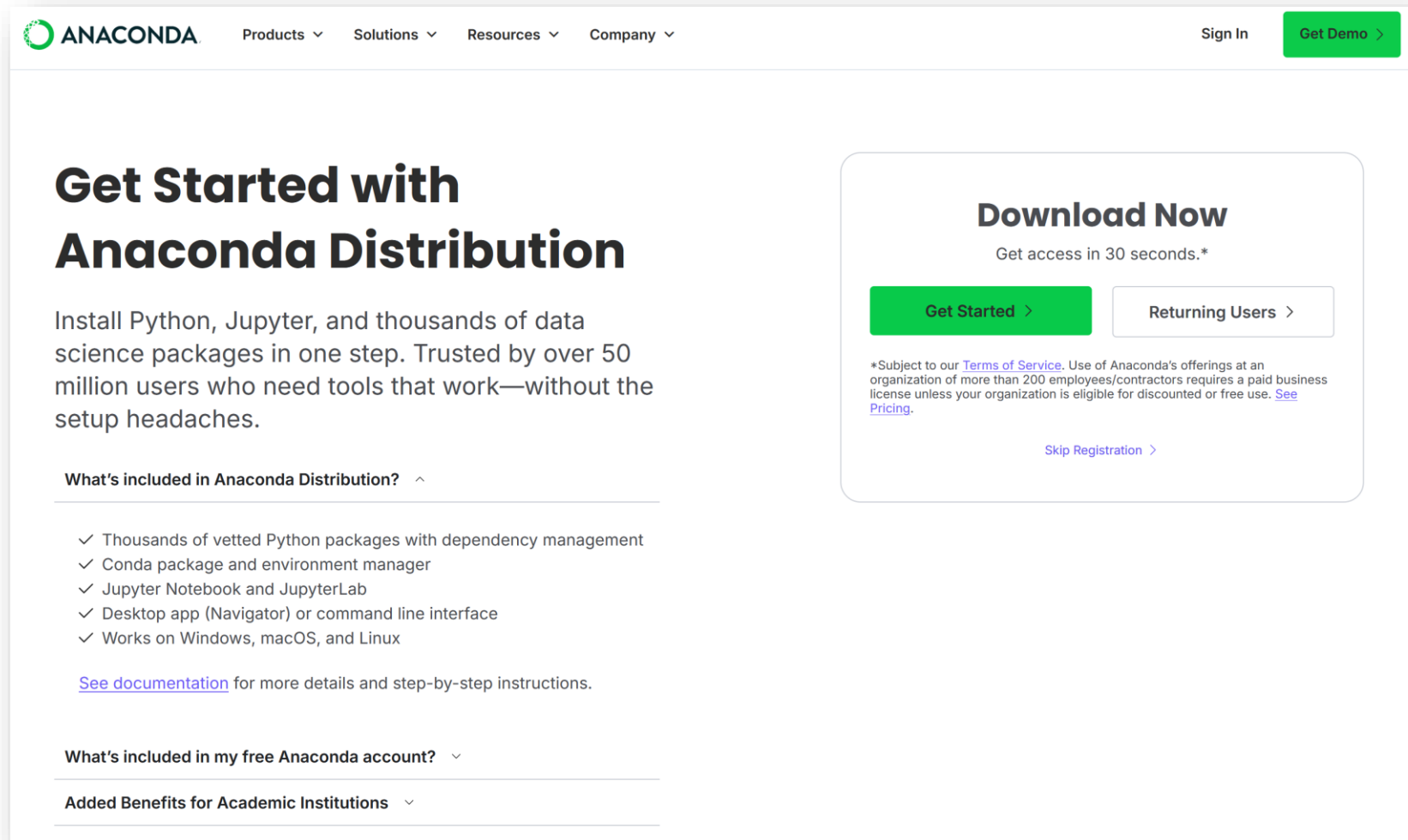
Any IDE can edit and run your scripts in any environment you created.

# Anaconda based environments

- Anaconda is easy to setup and could be used in different courses, projects and labworks (AI, Python course, M1 projects, etc.).
- Each Python project can have its own dependencies (libraries or packages) and its own Python version, regardless of what dependencies every other project has.
- It allows to create a separate *virtual* environment and a specific configuration for each different context.
- This tutorial is just an example. A lot of alternative solutions exist.
- This tutorial doesn't include GPU setup.

# 1. Download Anaconda

<https://www.anaconda.com/download/>



The screenshot shows the Anaconda website's download page. The header includes the Anaconda logo, navigation links for Products, Solutions, Resources, and Company, and buttons for Sign In and Get Demo. The main content area is divided into two columns. The left column features a large heading 'Get Started with Anaconda Distribution', a paragraph about installing Python, Jupyter, and data science packages, and a list of features included in the distribution. The right column has a 'Download Now' section with a 'Get Started' button and a 'Returning Users' button. Below these buttons is a disclaimer about terms of service and pricing, and a 'Skip Registration' link. The bottom of the page has sections for 'What's included in my free Anaconda account?' and 'Added Benefits for Academic Institutions'.

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## Get Started with Anaconda Distribution

Install Python, Jupyter, and thousands of data science packages in one step. Trusted by over 50 million users who need tools that work—without the setup headaches.

**What's included in Anaconda Distribution?** ^

- ✓ Thousands of vetted Python packages with dependency management
- ✓ Conda package and environment manager
- ✓ Jupyter Notebook and JupyterLab
- ✓ Desktop app (Navigator) or command line interface
- ✓ Works on Windows, macOS, and Linux

[See documentation](#) for more details and step-by-step instructions.

**What's included in my free Anaconda account?** v

**Added Benefits for Academic Institutions** v

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\*Subject to our [Terms of Service](#). Use of Anaconda's offerings at an organization of more than 200 employees/contractors requires a paid business license unless your organization is eligible for discounted or free use. [See Pricing](#).

[Skip Registration >](#)

# 1. Or miniconda

<https://docs.anaconda.com/miniconda/>

## Anaconda Distribution

Package and environment management with conda, Anaconda Navigator desktop app, and 600+ packages. Everything you need for data science.

### GRAPHICAL INSTALLER

↓ [Windows 64-Bit Graphical Installer](#)

📘 Graphical install instructions

### COMMAND LINE

📖 CLI install instructions

## Miniconda

Lightweight installer that only includes conda, Python, and their dependencies. Install only what you need.

### GRAPHICAL INSTALLER

↓ [Windows 64-Bit Graphical Installer](#)

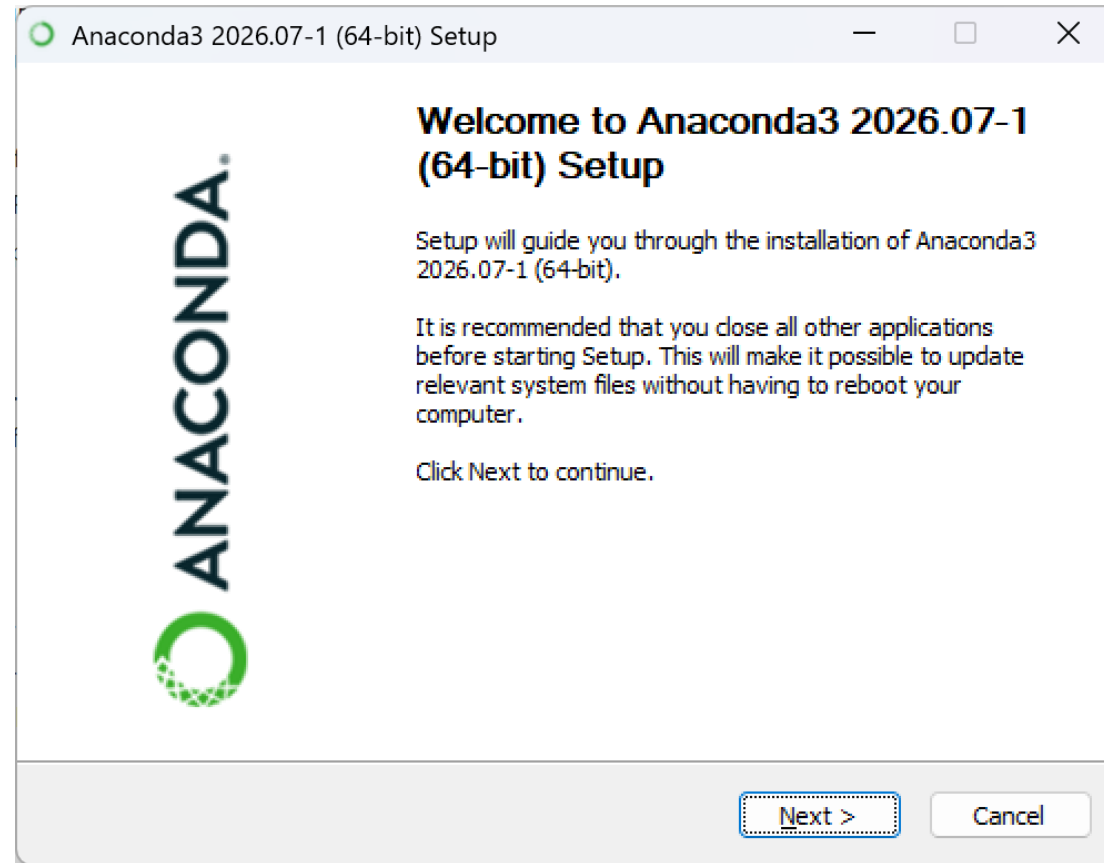
📘 Graphical install instructions

### COMMAND LINE

📖 CLI install instructions

Need help choosing an installer? [Compare Anaconda Distribution vs. Miniconda →](#)

## 2. Install Anaconda (or Miniconda)



Remember : **Anaconda = Python + Conda + A set of pre-installed useful libraries**

# 3. Update Conda

(and its packages, to the latest versions)

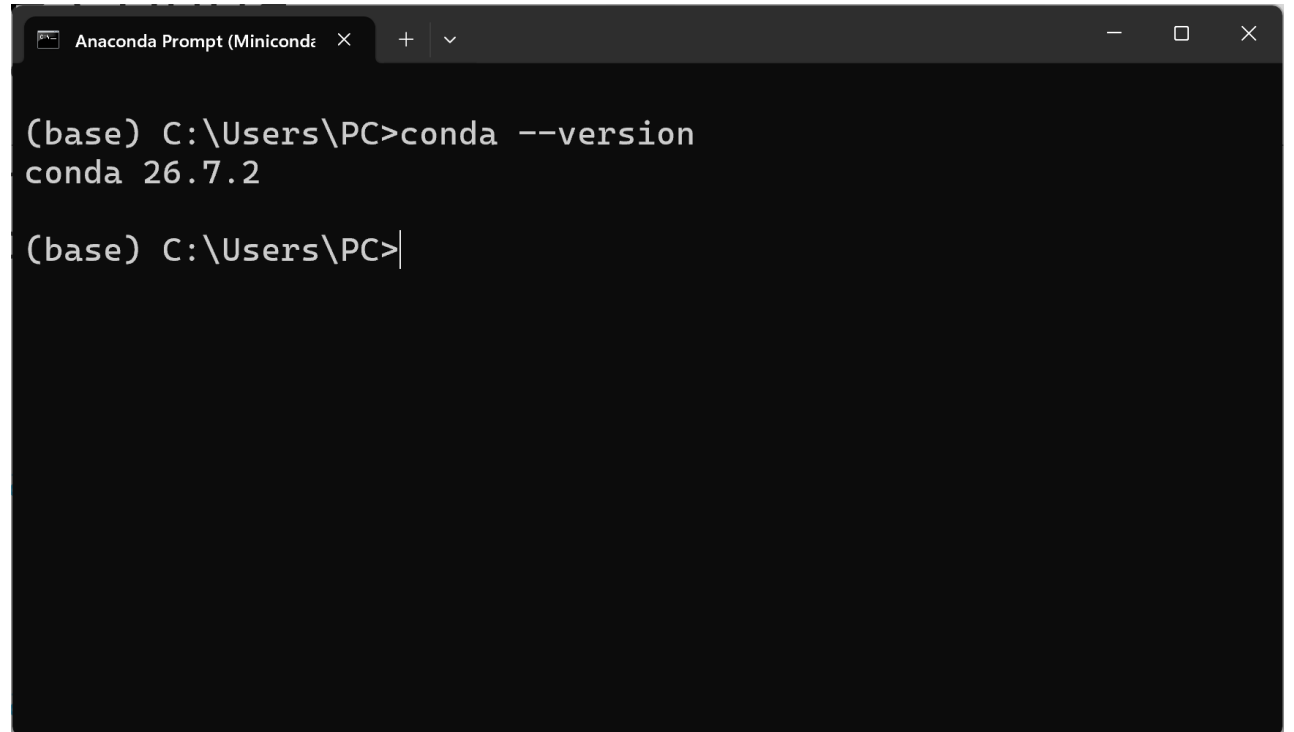
- Open a terminal or a command prompt or launch Anaconda prompt.  
Ex. in Windows : +R, then type `cmd`

- Run :

`conda update conda`

and

`conda update --all`



```
Anaconda Prompt (Miniconda) x + v
(base) C:\Users\PC>conda --version
conda 26.7.2
(base) C:\Users\PC>
```

## 4. Create your environment

- Open a terminal or a command prompt.

Ex. in Windows : +R, then type `cmd`

- Run (replace `my_environment_name` with your favorite name)

```
conda create -n my_environment_name
```

## 5. Activate your environment

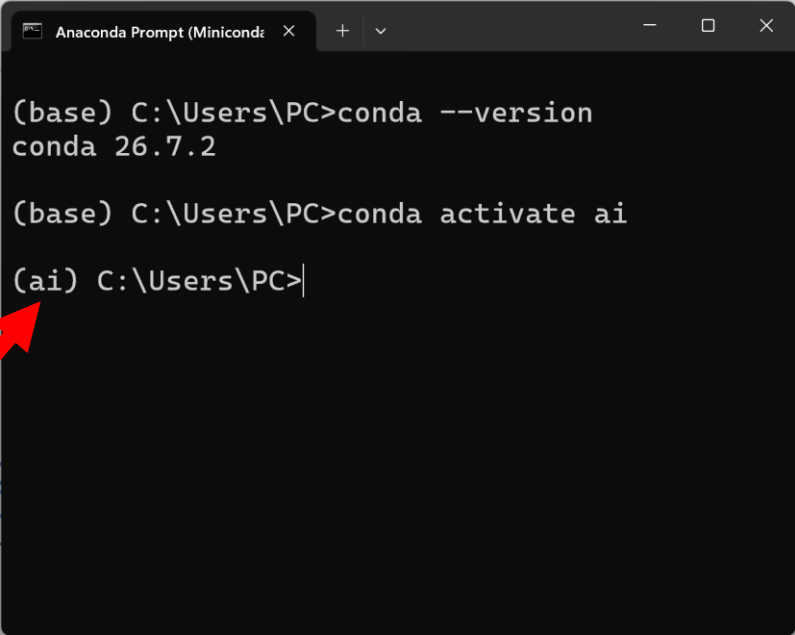
`activate my_environment_name`

- Or deactivate it :

`deactivate`

- Or test other Conda commands :

<https://docs.conda.io/projects/conda/en/4.6.0/downloads/52a95608c49671267e40c689e0bc00ca/conda-cheatsheet.pdf>



```
Anaconda Prompt (Miniconda) x + v - □ x
(base) C:\Users\PC>conda --version
conda 26.7.2

(base) C:\Users\PC>conda activate ai
(ai) C:\Users\PC>
```

## 6. Install librairies

- Installed librairies after this step will be available the activated environnement only

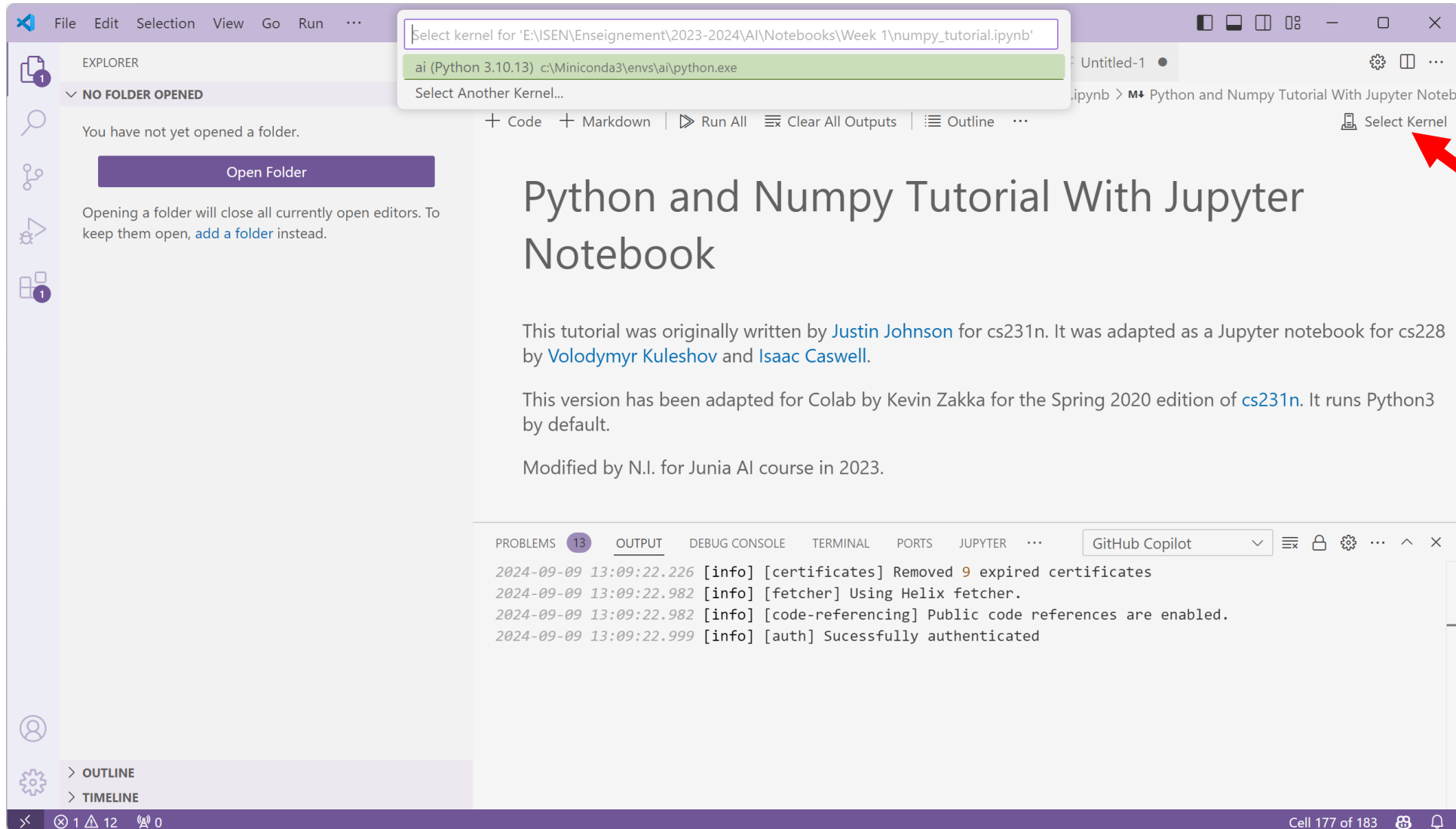
Install Jupyter      `conda install jupyter`

Run Jupyter        `jupyter-notebook`

Install numpy       `conda install numpy`

...

# 7. Test in VSCode



# Or use cloud solutions...

- If you **do not have a powerful PC**
- Or If you were **not able to run** all the previous steps
- Or you have a **problem with your setup** (Especially during exams)
- Or if you want just **a rapid Python/Ai playground**

**Use Google Colab:**

<https://colab.research.google.com/>

# Useful urls

<https://www.python.org/>

<https://www.anaconda.com/>

<https://docs.conda.io/projects/conda/en/latest/user-guide/tasks/manage-environments.html>

<https://developer.nvidia.com/cuda-downloads>

<https://developer.nvidia.com/cudnn>

[https://www.youtube.com/watch?v=82KLS2C\\_gNQ&list=PLO\\_fdPEVIfKqMDNmCFzQISI2H\\_nJcEDJq](https://www.youtube.com/watch?v=82KLS2C_gNQ&list=PLO_fdPEVIfKqMDNmCFzQISI2H_nJcEDJq)

See you Wednesday!

(Or next Monday!)